

AI Exploration Course: Entering the World of Artificial Intelligence

Build Your Very First Artificial Intelligence Project

Session 9 | A Practical Guide to Building Intelligent Tools Without Coding

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Session 9 Roadmap

Recap of Previous Session & Today's Objectives

Session 8 Recap: Ethics, Challenges, and Risks of Artificial Intelligence

Understanding algorithmic bias, deepfake detection, security vulnerabilities, and responsible usage principles.

Practical Objectives for Today's Session:

Going Beyond Coding: How can we build real, functional projects without programming knowledge?

Exploring No-Code Platforms: Getting to know Google AI Studio, Hugging Face, Make, and n8n

Examining 3 Practical Projects: Building a chatbot, an image classifier, and a content creation assistant

Step-by-Step Execution: Turning a simple idea into an interactive, presentable application

The No-Code Revolution

Why Can Anyone Build AI Projects Without Coding?

1. Assembling Pre-Built Lego Bricks (Visual Building Blocks)

Sophisticated models are pre-trained by tech giants; we simply connect them like Lego pieces to craft solutions.

2. Natural Language as the New Universal Programming Language

Instead of writing hundreds of lines of code, program logic and rules are defined through clear System Prompts.

3. Focusing on Problem Solving Over Technical Overhead

Transform an idea into a working prototype in under an hour without needing an engineering team.

Where Do We Begin?

Three Simple & Exciting Ideas for Your First Project

1. Specialized Chatbot & Knowledge Q&A Assistant

Build an assistant that accurately answers customer or student inquiries based on your reference documents.

2. Smart Image or Audio Classifier

A computer vision tool that identifies object types, plant health, or facial expressions from photos or webcams.

3. Automated Content Generation & Repurposing Engine

A workflow that takes a topic or link and simultaneously drafts LinkedIn posts, Instagram captions, and emails.

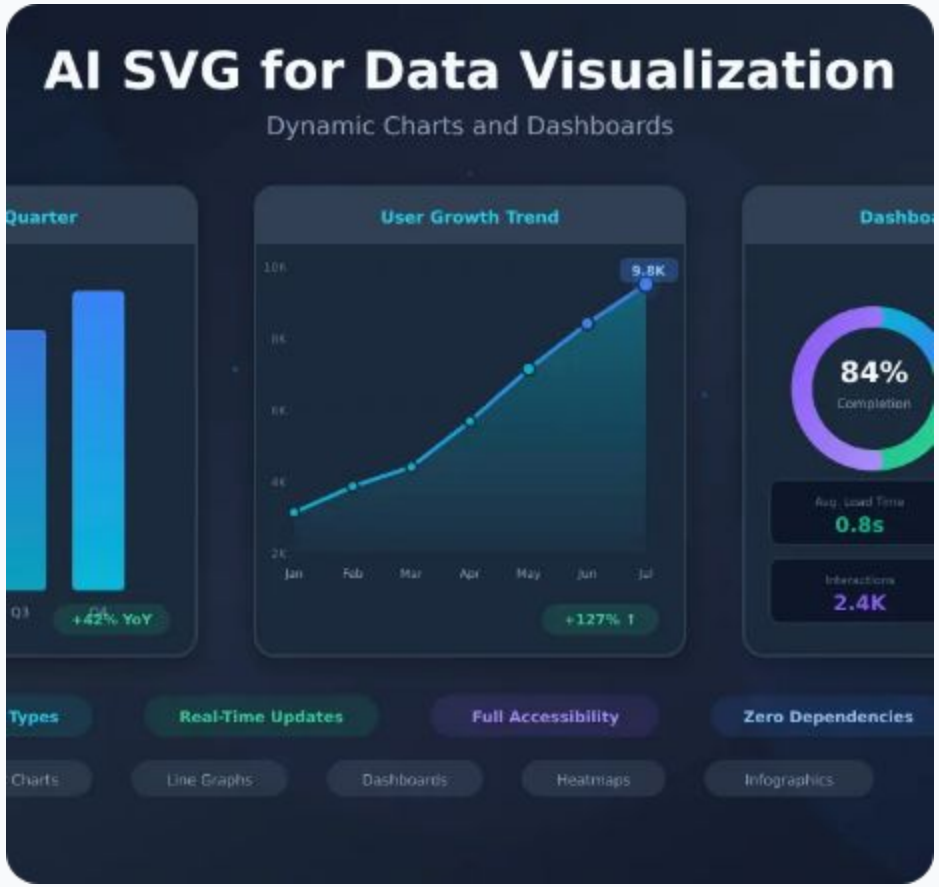
Builder's Toolkit

Comparing Top No-Code AI Platforms (From Beginner to Advanced)

Platform Name	Ease of Use for Beginners	Primary Project Use Case
Google AI Studio	Very Easy (Best Starting Point)	Rapidly create multimodal (text, audio, vision) assistants using Gemini models
Teachable Machine	Very Easy & Intuitive	Train custom in-browser image and audio recognition models in 5 minutes
Hugging Face Spaces	Intermediate	Explore thousands of community AI apps, test live demos, and duplicate projects
Make & n8n	Intermediate to Advanced	Connect AI logic to Telegram, Gmail, Google Sheets, and custom webhooks

Best Tool to Start With

Google AI Studio: Your Free AI Prototyping Sandbox



1. System Instructions (The Heart of Your Agent)

Define the persona, behavioral rules, tone, and primary task of your intelligent assistant in plain language.

2. Grounding with Custom Files & Documents

Upload PDFs, books, images, or videos so the model answers strictly based on your proprietary knowledge base.

3. Saving & Exporting API Keys

Save your crafted prompts and connect them seamlessly to external tools and applications with one click.

Global AI Showcase

Hugging Face Spaces: Thousands of Ready-to-Use Projects

1. The Free App Store of Machine Learning

A global hub where developers share interactive applications—from background removal to voice synthesis—for free.

2. Direct In-Browser Testing with Zero Installation

Simply explore the Spaces directory, open any public tool, provide your input, and see instant output.

3. Clone & Customize Any Space (Duplicate Space)

Duplicate any community project to your own profile with one click to adjust its parameters, branding, and logic.

Connecting AI to the Real World

Make & n8n: Seamlessly Integrating AI into Everyday Apps



1. Visual Drag-and-Drop Workflow Design

Connect Telegram or Gmail triggers to ChatGPT/Gemini and log outputs into Google Sheets without code.

2. Make Platform (Ideal for Rapid Start)

An intuitive, visually engaging interface with hundreds of pre-built templates for everyday automations.

3. n8n Platform (Powerful Agentic Workflows)

An advanced automation platform perfect for building AI agents equipped with memory and search tools.

Sample Project 1

Project 1: Building a Support or Educational Chatbot



Recommended Tool: Google AI Studio (or Custom GPTs)

Step 1: Write System Instructions

“You are an academic admissions advisor; answer politely and strictly based on the attached reference document.”

Step 2: Attach Knowledge Files (Mini Knowledge Base)

Upload a PDF containing course catalogs, tuition fees, schedules, and frequently asked questions.

Step 3: Test & Share the Chatbot

Run test questions in the chat preview panel, adjust instructions, and export or share the completed project.

Sample Project 2 (Computer Vision)

Project 2: Build an Image Recognition Model in 10 Minutes

Recommended Tool: Teachable Machine (Free Google Product)

Project Concept: Classifying 'healthy vs. defective products' or 'healthy vs. diseased plant leaves'

Phase 1: Define Classes & Collect Samples (Gather)

Upload ~15 images for Class 1 (e.g. healthy fruit) and ~15 images for Class 2 (defective fruit), or snap with webcam.

Phase 2: Train the Model (Train Model)

In under 30 seconds, the neural network learns visual distinguishing patterns directly in your browser.

Phase 3: Live Testing & Exporting (Export)

Present a new item in front of your camera to inspect real-time classification confidence percentages!

Sample Project 3 (Automation)

Project 3: Automated Content Generation Factory

Recommended Setup: Google Sheets + Make Platform + LLM API

Objective: You enter only the topic in an Excel row; all remaining columns are generated automatically!

Node 1 (Input Trigger): Watch New Rows in Google Sheets

Whenever a new row is added (e.g. topic 'Benefits of Daily Walking'), the automation workflow triggers.

Node 2 (Processing): Content Drafting via Language Model

Simultaneously drafts an Instagram caption, 5 relevant hashtags, and a story hook following your custom template.

Node 3 (Output Destination): Write to Sheet or Post to Telegram

Saves generated copy directly back into adjacent spreadsheet columns or sends it to your Telegram channel.

Roadmap from Concept to Deployment

Four Golden Steps to Building Any AI Project

Step 1: Define a Concrete, Focused Problem (Problem)

Instead of trying to 'automate the entire company', target a specific goal like 'summarizing customer reviews'.

Step 2: Curate Clean Data & Test Samples (Data)

Prepare a clean text reference file, a few sample Q&A pairs, or clear photos relevant to the specific domain.

Step 3: Select Tool & Build V1 Prototype (Build)

Implement quickly using the simplest accessible tool (such as Google AI Studio or Teachable Machine).

Step 4: Test Edge Cases & Refine Prompts (Test & Refine)

Evaluate against challenging inputs, establish output guardrails, and fine-tune system instructions.

Creator's Secrets to Success

Key Takeaways When Building Your First Project

1. Keep It Simple & Start Small

A simple working project that solves one real need is a thousand times more valuable than a complex concept on paper.

2. Establish Clear Guardrails for Chatbots

Explicitly state in system instructions: 'If the answer is not in the source text, state you do not know and do not hallucinate.'

3. Leverage Few-Shot Demonstration Examples

Providing 2–3 examples of user questions paired with ideal responses dramatically improves response accuracy.

Live Hands-on Classroom Workshop

Take Action: Build Your First AI Assistant in 15 Minutes!

Step-by-Step Exercise with the Instructor (in Google AI Studio):

Goal: Build an 'Intelligent Resume Critique Assistant' or a 'Technical Jargon Plain-English Translator'.

Minutes 1 to 5: Open `aistudio.google.com`, log in with Google, and click 'Create Prompt'.

Minutes 5 to 10: Define agent behavior, tone, and guardrails in the System Instructions box.

Minutes 10 to 15: Test the assistant with 3 realistic inputs and save your working prompt.

Presentation & Showcasing Skills

How to Professionally Present Your AI Projects

1. Three-Part Storytelling Formula (Problem ➔ Solution ➔ Impact)

Start with the pain point addressed, introduce your AI solution, and highlight measurable time or cost saved.

2. Live Interactive Demo or 1-Minute Screen Recording

Instead of theoretical slides, demonstrate live input submission and show the impressive output in action.

3. Publish on LinkedIn & Portfolio

Share project screenshots, workflow summaries, or live links as portfolio proof of practical skills.

Final Review

Session 9 Key Takeaways

1. Building is Accessible to Everyone: With modern no-code tools, anyone can build functional AI apps without writing code.

2. Select the Right Tool for Your Goal: Use Google AI Studio for chatbots, Teachable Machine for vision, and Make/n8n for automation.

3. Start Small and Iterate: Solve one focused, real problem first, then expand capabilities gradually over time.

4. High Value of Showcasing Your Work: Documenting and presenting your projects builds an invaluable portfolio for professional growth.

Q&A & Clarifications

Frequently Asked Questions About No-Code AI

Are Google AI Studio and Teachable Machine completely free to use?

Yes, both tools are provided free by Google for learning, testing, and rapid prototyping.

Do no-code AI projects hold genuine value in professional workplaces?

Absolutely! Many modern companies use Make, n8n, and Google AI Studio to automate workflows and build internal tools.

What should we do if we encounter errors while building a project?

Take a screenshot of the error and share it with ChatGPT or Claude; they will guide you step-by-step through the fix!

Practical Course Project

Session 9 Assignment: Build & Present Your First Mini-Project!

Action Mission: Transform one of today's 3 project concepts into a working solution

Choose a topic related to your personal interests, studies, or career, and build it using one of today's tools.

1. **Select & Build:** Build an assistant in Google AI Studio, a classifier in Teachable Machine, or an automation flow.

2. **Simple Documentation:** Capture 2 screenshots (one of your configuration/prompt and one showing the working result).

3. **Brief Summary:** Write a 3-line summary: Project Goal, Tools Used, and Practical Impact.

End of Session 9 | AI Exploration Course

Thank you for your active participation and passion for building!

Next Session (Session 10): Course Wrap-up, Project Presentations, and Your Future AI Roadmap

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Image Sources

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Source: edcircuit.com



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Source: www.xainflow.com



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